# "I built C++ primarily for myself and my colleagues"

When we think about programming, it's C++ that first comes to mind. That's because of the immense popularity and broad applicability of the general-purpose programming language that debuted back in 1983. Since then, it has influenced many modern languages, including C#, D and even Java. But what makes C++ important in the world of generation-next computers and devices like smartphones and embedded hardware? ***Jagmeet Singh*** of ***OSFY*** tries to get the answer to this question in a conversation with none other than the ***creator of C++, Bjarne Stroustrup.*** Edited excerpts...

**Q What prompted you to develop C++?**

I needed a language that could be used to manipulate hardware directly and use all the available hardware resources well. I also needed a language that allowed me to handle complexity. Though C could be used in manipulating hardware and Simula could handle complexity, there wasn't a language that could do both. Therefore, I started to build one, by adding Simula's class ideas to C.

**Q For whom did you build the initial C++ model?**

I wanted that language for myself, to help build a distributed system based on UNIX. Before I even finished my language, my friends and colleagues at Bell Labs started to use it for their projects, often simulation projects because the very first library I built for 'C with Classes' (as I called my language) was a co-routine library.

I built C++ primarily for myself and my colleagues. However, the range of projects, the demands and the hardware at Bell Labs were very wide, so the language had to be very flexible to cope with all of that.

**Q What were the prime challenges you faced when building C++?**

There were many challenges because I was building a tool for practical use, rather than as an academic project for publication. A tool has to be good enough for everything its users need; just being the best in the world for one or two things is not sufficient for success.

The very first challenge that I faced was the language's design. The question that arose was what features do my colleagues and I need in order to simplify our code. Implementing those features so that they were affordable to use in real-world development and execution environments was also quite hard, initially.

Once the language was ready, its installation and portability on a variety of hardware and operating systems was quite tough. Similarly, educating people on how to use the new techniques and language features was also a big challenge.

It was not all that easy because for the first years I was the only person working on 'C with Classes'. My colleagues were most supportive, but there was no official C++ project with a budget; basically, the help I offered to a range of Bell Labs projects allowed me to develop C++.

**Q Why was there a need for C++ when C already existed in the computing world?**

I built C++ on C because I did not want to build from scratch. That would not have resulted in a useful tool within a reasonable time frame. However, C could not, and still cannot, manage complexity as well as C++. The C language, and how it is used,